Class 2, Week 9: Binary classification (yes or no) and how to measure it

Lecturer guide for CML3112, CML3113 and CML3111. Keep it private.

The class in one paragraph

Can a binary classifier say yes or no to "high cooling load", and how do we count its true and false positives and negatives?

Welcome to Class 2. Last week the model gave you a number. Today it answers yes or no, which is called binary classification. We stay with the building data, and the question is whether the cooling load reaches its limit. You will see how a classifier is built, and the most important lesson of the day: how to count its mistakes. The ideas come first, then the lab, then a short quiz.

When each cohort meets

Cohort	Programmes	Date	Time and room
CML3112	Civil Engineering and Architecture	Tuesday 10 November 2026	11:00 to 14:00, Room 106
CML3113	Mining and Petroleum Engineering	Friday 13 November 2026	08:00 to 11:00, Room 105
CML3111	Electrical, Mechatronics, Robotics and Communications	Friday 13 November 2026	11:00 to 14:00, Room 115

Before the class

- Open the deck in Slide Show mode and the lab notebook in Colab before anyone arrives. Run the notebook from the top once on your account.
- Put the lab notebook and engineering_data.csv in the shared class folder and post the link. Every learner needs both files in Colab, in the Files panel.
- Write the "check your numbers" table on the board before the lab (the numbers for each cohort are in the table below).
- Bring the Block 1 quiz paper. The quiz opens in the last 20 minutes of this class, stays open for 24 hours, has five questions and covers Class 1 and Class 2. Learners sit it in class. The paper itself is not part of this deck.
- Check the deadline and the form of the assignment before you confirm them in writing. They are not decided in this guide.

The clock

Tuesday cohort (CML3112): the first full teaching of this class. Friday cohorts (CML3113, CML3111): the same deck and the same clock as the Tuesday cohort, because the concepts of this class are new to all three cohorts.

CML3112, Tuesday

Starts	Minutes	What happens
0:00	10	Recall from Class 1
0:10	35	Concepts 1: labels, z-scores, sigmoid
0:45	30	Concepts 2: confusion matrix, thresholds
1:15	15	Break
1:30	70	Lab: binary classification, in the notebook
2:40	15	Quiz opens: five questions, open 24 hours

Starts	Minutes	What happens
2:55	5	Close and what is assessed

CML3113 and CML3111, Friday

Starts	Minutes	What happens
0:00	10	Recall from Class 1
0:10	35	Concepts 1: labels, z-scores, sigmoid
0:45	30	Concepts 2: confusion matrix, thresholds
1:15	15	Break
1:30	70	Lab: binary classification, in the notebook
2:40	15	Quiz opens: five questions, open 24 hours
2:55	5	Close and what is assessed

Slide map

The deck has two halves. Slides up to the divider are concepts: no laptops. After the divider come the lab slides and the notebook. You switch once.

Slide	Title	Part	Friday recap
1	Binary classification (yes or no) and how to measure it	Concepts	
2	Today in one view: what you will leave with	Concepts	
3	Your three hours: ideas first, then the lab, then the quiz	Concepts	
4	Three questions from last time	Concepts	
5	Answers to the three questions from Class 1	Concepts	
6	Words we use today	Concepts	
7	Binary classification gives a yes or no answer that tells a plant operator whether to act	Concepts	
8	Regression predicts a number and classification predicts a class	Concepts	
9	A limit on the cooling load turns each row into a positive or negative label	Concepts	
10	The majority-class baseline always answers no, is right on 80 percent of rows and finds no actual positive	Concepts	
11	Standardisation (StandardScaler) puts load_pct and ambient_c on the same scale before the model learns	Concepts	
12	The z-score formula: subtract the training mean, then divide by the training standard deviation	Concepts	
13	Logistic regression passes a linear score through the sigmoid (logistic) function, which stays between 0 and 1	Concepts	
14	Logistic regression in three lines: score, probability, prediction	Concepts	
15	Worked example: validation row 3 gets p = 0.45, so the model predicts no	Concepts	
16	A decision tree classifier can also answer yes or no, and it does not need standardisation	Concepts	
17	How to read the tree	Concepts	

Slide	Title	Part	Friday recap
18	The confusion matrix counts true and false positives and negatives	Concepts	
19	Accuracy and precision are worked out from the four counts TP, FP, FN and TN	Concepts	
20	Recall (sensitivity) and F1: how many actual positives we find, and one blended score	Concepts	
21	The four scores in words: the question each one answers	Concepts	
22	A false negative and a false positive cost different things for a building	Concepts	
23	Logistic regression and the decision tree give different counts on the same 15 rows	Concepts	
24	Moving the decision threshold changes the precision-recall trade-off: more positives found, more false positives	Concepts	
25	Each decision threshold gives its own counts and its own scores	Concepts	
26	On a larger set, a lower decision threshold finds more positives and also gives more false positives	Concepts	
27	Two rates give the points of the ROC curve: the false positive rate (FPR) and the true positive rate (TPR)	Concepts	
28	The ROC curve plots one point for every decision threshold: FPR across, TPR up	Concepts	
29	AUC is the area under the ROC curve, and on a large dataset the curve is smooth	Concepts	
30	With 15 rows and 3 actual positives, every score here is shaky	Concepts	
31	Summary: judge a binary classifier by counting each kind of mistake	Concepts	
32	Read the workflow before you write it: impute, standardise, fit, predict probabilities, count	Concepts	
33	Part 2: the lab	Lab	
34	The lab notebook has five stages, and each stage ends with a check	Lab	
35	Check your numbers against these before you move on	Lab	
36	Four messages you will meet, what they mean and what to do	Lab	
37	How this class counts towards your coursework	Lab	
38	Save the notebook and keep your scores	Lab	

Questions to ask, and a good answer

Ask	A good answer
What does a false positive cost for your equipment?	Answers differ. Listen for a cost in time or energy, and compare it with the cost of a false negative.
Why is accuracy alone not enough?	With few positive cases, the majority-class baseline scores 0.80 and finds none of them.
When is precision the more important score?	When a false positive is expensive, for example an unnecessary stop.
When is recall the more important score?	When a false negative is dangerous or expensive, for example an overheating motor.
What happens to recall if you lower the decision threshold?	It rises or stays the same. In the building data it goes from 0.67 at 0.5 to 1.00 at 0.3.

Ask	A good answer
Why do we fit StandardScaler on the training set only?	If the validation set helped decide the mean and the standard deviation, information would leak from held-out data into the model. That is data leakage.
Does the decision tree need standardisation?	No. A tree only asks whether a value is above or below a threshold, and standardisation does not change the order of the values.
What does an AUC of 0.5 mean?	The model ranks positive and negative cases no better than a coin flip.

Ideas learners often get wrong

What a learner may say	What to say back
High accuracy means a good model	On the validation set the majority-class baseline (always predict no) already scores 0.80. Compare with that baseline and look at recall.
The model says yes when the probability is high, so the decision threshold is always 0.5	0.5 is a default. The right decision threshold depends on the cost of a false negative and of a false positive.
Standardisation changes the answer of the model	It changes the units of the weights. The probabilities of this model stay almost the same, and the weights become comparable.
An AUC of 1.0 means the model is perfect	It means every actual positive scored above every actual negative in these 15 rows. New equipment can behave differently.

Walking the lab

Step	What to look for
Step 1: prepare	Rows after cleaning 88. Split 58, 15, 15. The same cleaning and split as Class 1, repeated.
Step 2: label	Training 19 of 58 are positive, validation 3 of 15. Majority-class baseline accuracy 0.80.
Step 3: logistic regression	Weights 2.79 and 0.73, intercept -1.71. The CHANGE task removes StandardScaler. The weights become small, because each weight now belongs to one percent of load or one degree of ambient temperature and no longer to one standard deviation. A ConvergenceWarning may appear.
Step 4: decision tree	Depth 3. TP 2, FN 1, FP 1.
Step 5: confusion matrix and scores	TP 2, FN 1, FP 0, TN 12. Accuracy 0.93, precision 1.00, recall 0.67, F1 0.80. Ask one learner to repeat one division at the board.
Step 6: decision thresholds	Lowering the decision threshold from 0.5 to 0.3: recall goes from 0.67 to 1.00, and false positives (FP) goes from 0 to 1. Raising it from 0.5 to 0.7: recall goes from 0.67 to 0.00, and false positives (FP) stays at 0. At 0.7 the model predicts positive for no row at all, so precision has nothing to divide and we show it as 0.00.
Step 7: ROC curve and AUC	AUC 1.00, which is 36 of 36 pairs in the right order. Ask what a score this high on 15 rows promises.
Steps 8 and 9: conclusion and save	Three sentences from every learner. Insist on restart and run all before saving.

The numbers each cohort should see

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Label limit	54 kW	13.5 kWh/t	80 deg C
Positive labels, training set	19 of 58	24 of 58	17 of 58
Positive labels, validation set	3 of 15	3 of 15	3 of 15

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Majority-class baseline accuracy	0.80	0.80	0.80
Logistic weights: load, ambient, intercept	2.79, 0.73, -1.71	2.64, 0.49, -0.95	2.81, 0.93, -1.99
Logistic: TP, FN, FP, TN	2, 1, 0, 12	3, 0, 1, 11	2, 1, 0, 12
Logistic: accuracy, precision, recall, F1	0.93, 1.00, 0.67, 0.80	0.93, 0.75, 1.00, 0.86	0.93, 1.00, 0.67, 0.80
Decision tree depth 3: TP, FN, FP, TN	2, 1, 1, 11	3, 0, 1, 11	2, 1, 0, 12
Decision tree depth 3: accuracy, precision, recall, F1	0.87, 0.67, 0.67, 0.67	0.93, 0.75, 1.00, 0.86	0.93, 1.00, 0.67, 0.80
Recall at decision thresholds 0.3, 0.5, 0.7	1.00, 0.67, 0.00	1.00, 1.00, 0.67	1.00, 0.67, 0.00
FP at decision thresholds 0.3, 0.5, 0.7	1, 0, 0	2, 1, 0	0, 0, 0
AUC	1.00	0.97	1.00

If things go wrong

- If the concepts run over, skip the slide that compares the two models and say the results in one sentence. Keep the confusion matrix, the four scores and the decision threshold slides.
- If only 45 minutes of lab remain, do steps 1 to 5 and the threshold step. Set the decision tree and the ROC curve as practice for next time.
- If the quiz time is under pressure, shorten the lab by dropping the decision tree. The quiz covers both classes and must still happen today.
- If wifi fails in the room, use the offline set-up from Week 2: the notebook runs unchanged in VS Code.
- If several learners are stuck at the same cell, stop the room, run that cell on the projector and let them copy the fix.

After the class

- Share the lab notebook and engineering_data.csv again so absent learners can follow.
- Confirm in writing the deadline and the form of the Block 1 assignment. Do not give a date before it is decided.
- Note any question you could not answer and start Class 3 with it.

Speaker script

Every slide carries its spoken script in the PowerPoint notes (open Presenter View). The scripts are first person and are meant to be read or paraphrased. Pauses are marked by questions.